240.1

$$J = \begin{pmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{pmatrix} \in \mathscr{M}_n(\mathbb{R})$$

$$\chi_J = \begin{vmatrix} X-1 & -1 & \dots & -1 \\ -1 & X-1 & \dots & \vdots \\ \vdots & & \ddots & \vdots \\ -1 & \dots & \dots & -1 \end{vmatrix}$$

$$= \begin{vmatrix} X-n & -1 & \dots & -1 \\ X-n & X-1 & \dots & \vdots \\ \vdots & & \ddots & \vdots \\ X-n & -1 & \dots & X-1 \end{vmatrix} C_1 \leftarrow C_1 + C_2 + \dots + C_n$$

$$= (X-n)\begin{vmatrix} 1 & -1 & \dots & -1 \\ \vdots & X-1 & & \vdots \\ \vdots & -1 & \ddots & \vdots \\ 1 & -1 & \dots & X-1 \end{vmatrix}$$

$$= (X-n)\begin{vmatrix} 1 & 0 & \dots & 0 \\ \vdots & X & & \vdots \\ \vdots & & \ddots & \vdots \\ 1 & \dots & \dots & X \end{vmatrix}$$

$$= (X-n)X^{n-1} \text{ car matrice triangulaire}$$

$$Sp(J) = \{0, n\}$$

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in E_0(J) \Leftrightarrow JX = 0$$

$$\Leftrightarrow \begin{cases} x_1 + \dots + x_n = 0 \\ x_1 \dots x_n = 0 \end{cases}$$

$$\Leftrightarrow x_1 + \dots + x_n = 0$$

$$\Leftrightarrow x_n = -x_1 - \dots - x_{n-1}$$

$$\Leftrightarrow X = \begin{pmatrix} x_1 \\ \vdots \\ -x_1 - \dots - x_{n-1} \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ -1 \end{pmatrix} + \dots + x_{n-1} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -1 \end{pmatrix} \Leftrightarrow X \in Vect(U_1, \dots, U_{n-1})$$

Donc $E_0(J) = Vect(U_1, \dots, U_{n-1})$
$E_0(J) = Ker(J - 0I_n)$

240.3

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \qquad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

a/

$$AM = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix} \qquad MA = \begin{pmatrix} a & 2b \\ c & 2d \end{pmatrix}$$

$$AM - MA = \begin{pmatrix} a & -b \\ c & 0 \end{pmatrix}$$

b/

$$\varphi : \mathscr{M}_2(\mathbb{R}) \to \mathbb{M}_2(\mathbb{R})$$
$$M \mapsto AM - MA$$

M2

$$M2 : AM - MA = \lambda M$$
$$\Leftrightarrow \begin{pmatrix} a & -b \\ c & 0 \end{pmatrix} = \lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
$$\Leftrightarrow \begin{cases} \lambda a = 0 \\ \lambda b = -b \\ \lambda c = c \\ \lambda d = 0 \end{cases}$$

1er cas $\Leftrightarrow \lambda = 0$ $\begin{cases} b = 0 \\ c = 0 \end{cases}$
Donc $E_0(\varphi) = \{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} | a, d \in \mathbb{R} \}$
Si $\lambda = 1$ a = b = d = 0
$E_1(\varphi) = \{ \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} | c \in \mathbb{R} \}$
Si $\lambda = -1$
$E_{-1}(\varphi) = \{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} | b \in \mathbb{R} \}$

240.2

$$\chi_c = \begin{vmatrix} X & 0 & \dots & 0 & a_1 \\ -1 & X & & & \vdots \\ 0 & -1 & \ddots & & \vdots \\ \vdots & & \ddots & X & \vdots \\ 0 & \dots & 0 & -1 & -a_{n-1} \end{vmatrix} \quad L_1 \leftarrow L_1 + XL_2 + X^2L_3 + \dots {}_X^{n-1} L_n$$
$$= \begin{vmatrix} 0 & \dots & \dots & \dots & 0 & P \\ -1 & X & & & & a_1 \\ 0 & -1 & X & & & \vdots \\ & & \ddots & \ddots & & \vdots \\ & & & \ddots & X & a_{n-2} \\ 0 & & & 0 & -1 & X + a_{n-1} \end{vmatrix}$$
$$P = a_0 + Xa_1 + X^2a_2 + \dots + X^{n-1}(X + a_{n-1})$$
dev $L_1 = (-1)^{n+1} P(-1)^{n-1} = P$

b/

$$V = V(\lambda_1, \dots, \lambda_n) = \begin{pmatrix} 1 & \dots & 1 \\ \lambda_1 & \dots & \lambda_n \\ \vdots & & \vdots \\ \lambda_1^{n-1} & \dots & \lambda_n^{n-1} \end{pmatrix}$$
$$det(V(\lambda_1, \dots, \lambda_n)) = \prod_{1 \le i < j \le n} (\lambda_j - \lambda_i) \neq 0$$

Donc V^{-1} existe
B = $C_1 \dots C_n$ avec $V = (C_1 | \dots | C_n)$

V = Pass($\mathscr{C} \to B$)

$$u(C_k) = C^T C_k$$

$$= \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & 0 & 1 \\ -a_0 & \dots & \dots & -a_{n-1} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ \vdots \\ \lambda_k^{n-1} \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_k \\ \lambda_k^2 \\ \vdots \\ \lambda_k^{n-1} \\ -a_0 - a_1\lambda_k - a_2\lambda_k^2 - \dots - a_{n-1}\lambda_k^{n-1} \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_k \\ \lambda_k^2 \\ \vdots \\ \lambda_k^{n-1} \\ -P(\lambda_k) + \lambda_k^n \end{pmatrix}$$

$$= \lambda_k \begin{pmatrix} 1 \\ \vdots \\ \lambda_k^{n-1} \end{pmatrix} \qquad = \lambda_k C_k$$

donc

$$Mat(u, B) = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

240.6

$$\chi_J = X^n - 1$$

χ_J est scindé donc J diagonalisable
b/ u canoniquement associé à J dans $\mathscr{B}(e_0, \dots, e_{n-1})$

$$u(e_j) = e_{j-1[n]}$$

donc $u^k(e_j) = e_{j-k[n]}$
J^k = J où les colonnes sont décalées de k places modulo n a droite

$$A = A_0 I_n + a_1 J + \dots + a_{n-1} J^{n-1}$$
$$= P(J) \text{ où } P = a_0 + a_1 X + \dots + a_{n-1} J^{n-1}$$

où $J = Q D Q^{-1}$ avec

$$D = \begin{pmatrix} 1 & & & & \\ & \omega & & & \\ & & \omega^2 & & \\ & & & \ddots & \\ & & & & \omega^{n-1} \end{pmatrix}$$

$$J = Q \begin{pmatrix} P(1) & & & & & \\ & P(\omega) & & & & \\ & & P(\omega^2) & & & \\ & & & \ddots & & \\ & & & & P(\omega^{n-1}) \end{pmatrix} Q^{-1}$$

donc

$$detA = \prod_{k=0}^{n-1}(\sum_{i=0}^{n-1} a_i\omega^{ki})$$

240.4

$$A = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$
$$Sp(A) = \{0, 2, 6\}$$
$$u(e_1) = 6e_1$$

Donc $e_1 \in E_6(u)$ or $dimE_6(u) \leq m_6 \leq 1$
Donc $E_6(u) = Vect(e_1)$
de même $E_2(u) = Vect(e_2)$
$u(e_2 + e_3) = 0$ et $dimE_0(u) = 1$
Donc $E_0(u) = Vect(e_2 + e_3)$
On pose $D = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} = Mat(u, B')$ et P $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = Pass(B \to B')$ par changement de base $\boxed{A = PDP^{-1}}$ c/
(E) $X^2 + X = A$ avec $X \in \mathscr{M}_3(\mathbb{R})$
Analyse
Soit X tq $X^2 + X = A$

$$\begin{aligned} XA &= X(X^2 + X) \\ &= X^3 + X^2 \\ &= (X^2 + X)X \\ &= AX \end{aligned}$$

$X^2 + X = A$ donc $PY^2P^{-1} + PYP^{-1} = PDP^{-1}$
donc $Y^2 + Y = D$

$$Y = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{pmatrix}$$
$$\text{donc } \begin{pmatrix} \alpha^2 & 0 & 0 \\ 0 & \beta^2 & 0 \\ 0 & 0 & \gamma^2 \end{pmatrix} + \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\text{donc } \begin{cases} \alpha^2 + \alpha - 6 = 0 \\ \beta^2 + \beta - 2 = 0 \\ \gamma^2 + \gamma = 0 \end{cases} \quad \text{cad } \begin{cases} (\alpha - 2)(\alpha + 3) = 0 \\ (\beta - 1)(\beta + 2) = 0 \\ \gamma(\gamma + 1) = 0 \end{cases}$$

240.31

$$A \in \mathscr{M}_n(\mathscr{K}), \exists p \in \mathbb{N}, A^p = 0_n$$

Mq $Sp(A) \subset \{0\}$
Soit λ vp de A

$$\begin{aligned} \exists X \neq 0_{n,1}, AX &= \lambda X \\ A^pX &= \lambda^p X \\ \lambda^p X &= \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \text{ Donc } \lambda = 0 \\ \chi_A &= X^n \\ \chi_A &= X^n - tr(A)X^{n-1} + ... + (-1)^n det(A) \\ trA &= 0 \end{aligned}$$

2/

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & 0 & 2 \end{pmatrix}$$

$$\begin{aligned} tr(A - \lambda I d_3) &= trA - \lambda tr(I d_3) \\ &= 6 - 3\lambda \end{aligned}$$

nécessairement $\lambda = 2$

Mq $B = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 2 & 0 & 0 \end{pmatrix}$ est nilpotente $B^3 = 0$

250.7

a/ Mq $P(A^T) = P(A)^T$

$$P = \sum_{k-0}^{n} a_k X^k$$

$$P(A^T) = \sum_{k=0}^{n} a_k (A^T)^k$$

$$P(A)^T = (\sum_{k-0}^{n} a_k X^k)^T = \sum_{k=0}^{n} a_k (A^k)^T$$

Mq : $(A^T)^k = (A^k)^T$

Relation de récurrence : $(A^T)^{k+1} = (A^T)^k A^T = (AA^k)^T$

b/ Contrexemple / vrai